

Sharetea SaaS MVP User Study: Accessibility-Focused Evaluation of Multi-View Interfaces

Executive Summary

This report presents the results of a structured user study conducted on the Sprint 1 Minimum Viable Product (MVP) of the Sharetea SaaS system, accessible at <https://project3-gangx1.vercel.app>. The MVP delivers the first integrated version of the customer kiosk, cashier interface, manager interface, and menu board views required for Project 3. Although feature scope at this stage is intentionally limited, the deployment establishes the base structure, layout system, visual style, and navigation patterns that future sprints will expand. The objective of this user study was to evaluate the clarity, usability, and accessibility of the MVP by observing how a real user—distinct from all team members—interacted with the system with no prior guidance.

The study followed a three-part method: (1) think-aloud interaction observation, (2) semi-structured user interview, and (3) a short targeted survey with both Likert-scale and open-ended responses. The goal was not to test feature completeness, but to identify friction points in navigation, information hierarchy, readability, interactive affordances, and overall user comprehension. These observations were then interpreted through the lens of three required accessibility personas: Maria, a bilingual user who needs clear language and reliable translation; Vishnu, a low-vision user dependent on magnification and high contrast; and Carol, a user with mild tremors and emerging visual impairment who relies on large targets and readable layouts.

The participant struggled to identify which elements were interactive because buttons and drink cards resembled static images. He also hesitated at the initial screen because the starting point for ordering was not obvious. Text was viewed as small, low-contrast, and visually understated, making readability difficult even for a fully sighted user. Click targets required precision, and spacing between elements was tight enough that mis-clicks occurred. Feedback during interaction was limited: clicking cards did not consistently produce noticeable state changes, hover states were subtle, and navigation transitions lacked strong visual confirmation. These issues collectively indicate that, at its current stage, the MVP establishes structural direction but requires substantial refinement in accessibility and interaction clarity.

Across both observation and interview components, several themes emerged: insufficient visual hierarchy, unclear navigation flow, small interactive elements, low color contrast, inconsistent spacing, and lack of feedback animations. These shortcomings are typical for a first-sprint deployment but are especially consequential when viewed through accessibility personas. Users like Maria need explicit labels and

simple language; Vishnu relies on contrast and scalable text; Carol needs oversized, widely spaced controls. The Sprint 1 MVP does not yet meet their needs, making accessibility a priority for Sprint 2.

This report concludes with a set of proposed design moves aligned with Project 3 requirements, WCAG 2.1 guidelines, and the constraints of the remaining sprints. These moves include strengthening text contrast, enlarging interactive elements, introducing explicit navigation labels, improving visual hierarchy, adding meaningful interaction feedback, and preparing stable UI locations for external APIs such as translation, authentication, and weather. Implementing these changes will bring the system closer to meeting accessibility expectations and provide a more intuitive, usable interface for all role-based views moving into Sprint 2.

Accessibility Personas

Maria requires simple language, minimal jargon, and clear task flows. She depends on machine translation and benefits from explicit labels, structured navigation, and consistent terminology. The kiosk interface must support bilingual clarity and avoid ambiguous icons.

Vishnu has low vision due to glaucoma, relying on magnification, large text, and strong contrast. He needs clear separation between sections, bold headings, visually distinct buttons, and layouts that do not collapse when zoomed.

Carol has macular degeneration and slight tremors, requiring large clickable areas, high-contrast typography, stable layout spacing, and components that do not require precision. She enlarges text in her browser and benefits from simple, uncluttered arrangement.

User Study Participant

The participant was “**Evan,**” (a name changed for privacy), a **21-year-old college junior**. Evan is comfortable with general technology but has no prior experience with kiosk-style bubble tea ordering systems. He does not have accessibility needs himself, which makes his friction points particularly useful because they reveal baseline issues that would be amplified for Maria, Vishnu, and Carol. He received only a description of the problem (a multi-view point-of-sale system) and was not given any explanation of the MVP before starting the think-aloud session.

User Study Findings

Evan’s experience revealed several consistent themes.

He hesitated at the landing view because he could not tell where to begin ordering. The interface lacked a clear “Start Order” or equivalent call-to-action. He scanned the page multiple times before deciding which

element to interact with, indicating insufficient navigation affordance. When browsing menu items, he described the drink cards as “pretty but not obviously buttons,” confirming that the interactive elements did not visually differentiate themselves from static content.

Text across the interface appeared small and low-contrast. Evan noted that he occasionally leaned closer to read item names or category labels. This is a critical barrier for low-vision users and directly affects both Vishnu and Carol. Several buttons and cards had minimal spacing between them, leading to inaccurate clicks even from a fully sighted, dexterous user.

State feedback was limited. Clicking a card produced little to no visual indication that an action occurred. Evan verbally expressed uncertainty whether the system had registered his click, demonstrating that clear interactive states should be prioritized. He also commented on the lack of guidance from one step to another, noting that the flow from browsing to ordering did not feel linear.

Evan did not notice any connection points for external APIs, such as translation or weather. Although this is expected at this stage, it signals the need for clearly reserved UI locations for these features in later sprints.

Proposed Design Moves

High-Level Design Move	Detailed Description	Justification
Improve contrast and enlarge text	Apply WCAG 2.1 AA contrast and increase base font size across kiosk, cashier, and menu board views	Supports Vishnu and Carol and improves general readability
Add explicit navigation labels	Add buttons such as “Start Order,” “Browse Menu,” and multilingual labels where appropriate	Supports Maria’s need for clarity and reduces confusion for all users
Increase size and spacing of click targets	Enlarge drink cards, buttons, and filters with generous spacing	Essential for Carol and prevents mis-clicks
Strengthen visual hierarchy	Add section headers, clearer grouping, and distinct card styles	Helps distinguish interactive elements
Add clear interactive feedback	Provide hover states, pressed states, and transitions	Helps users confirm actions, especially low-vision users
Reserve UI space for API integrations	Add visible translation toggle, weather display location, and Google authentication area	Aligns with required external API integration

Appendix: User Study Data

Interaction Observation

Task	Observation Summary	Insight
Start an order	User unsure where to begin; scanned screen repeatedly	Need strong call-to-action
Select a drink	User clicked several non-interactive areas	Interactive elements not visually obvious
Browse categories	Icons interpreted as decorations instead of buttons	Labels required
Attempt checkout	User unsure if drink was added	Need feedback animations

User Interview

Question	Summary of Response	Insight
What confused you most?	Could not tell what was clickable or where to start	Improve navigation cues
What worked well?	Visual style felt clean	Base aesthetic is strong
What felt missing?	Feedback animations and structured flow	Improve interaction states

User Survey

Question	Response	Insight
Navigation ease	3/5	Flow not intuitive
Text readability	2/5	Increase size & contrast
Button clickability	2/5	Targets too small
Interactivity clarity	2/5	Need stronger affordances
Additional comments	“Feels like an early prototype”	Confirms need for refinement